9 Strontium-90 is a radioactive isotope of strontium. The nuclide notation for strontium-90 is:

$$^{90}_{38}Sr$$

(a) (i) Explain what isotopes are.

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..... [1]

(ii) Complete Table 9.1 for strontium-90.

Table 9.1

particle	number in one atom	location
	38	outside nucleus
neutron		
	38	inside nucleus

[2]

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(b) Strontium-90 is used to measure the thickness of metal sheets in industry.
Strontium-90 decays by emitting beta (β) particles which pass through a metal sheet to a detector.

(i) One metal sheet is 0.75 mm thick. Suggest why strontium-90 is a suitable radioactive source to measure the thickness of the metal sheets.

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..... [2]

(ii) The half-life of strontium-90 is approximately 27 years. Fig. 9.1 shows the shape of a decay curve.

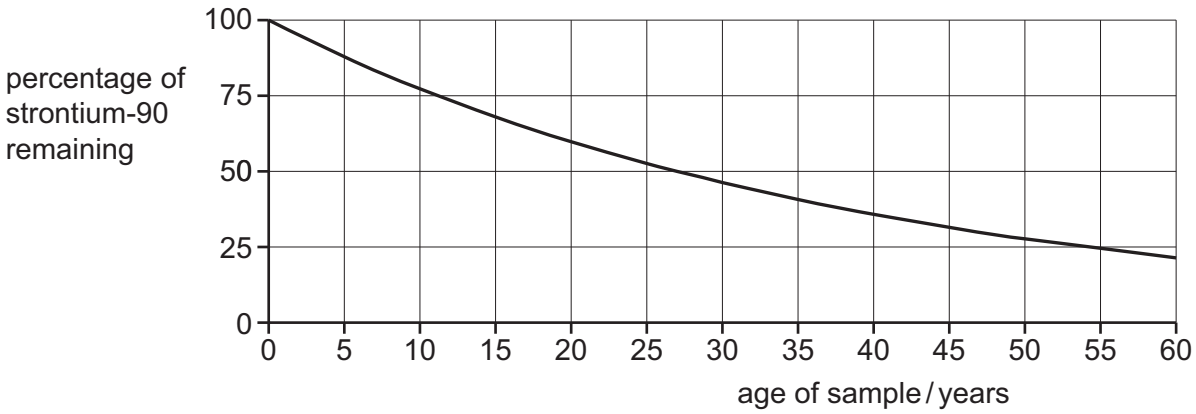

Fig. 9.1

The strontium-90 source is replaced with a new source after 15 years. Using Fig. 9.1, suggest why a strontium-90 source that is more than 15 years old needs to be replaced with a new source.

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